



LEEP SYSTEM 1000™ SERVICE MANUAL

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Distributed by:



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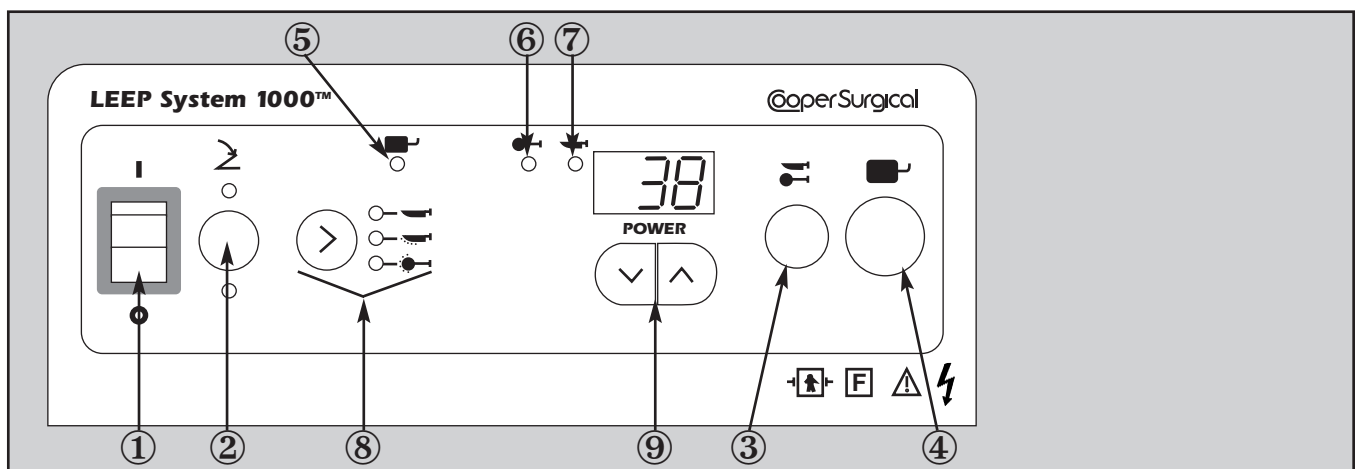
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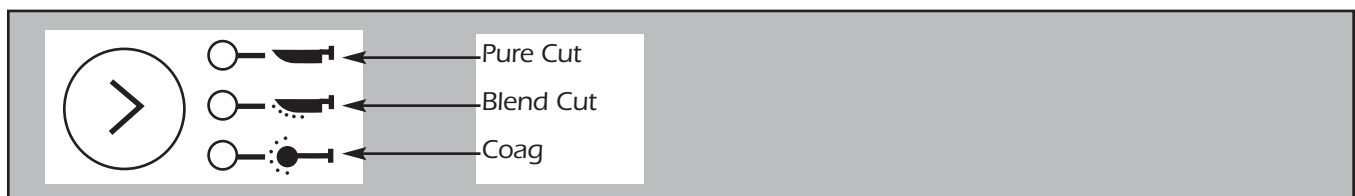
SYSTEM FEATURES

- Microprocessor controlled for increased precision, accuracy, repeatability, and safety.
- Adequate power for all LEEP™ monopolar electrosurgical procedures.
- Accurate selection of discrete power levels.
- Digital display of output power levels.
- Choice of radio frequency wave forms including CUT, BLEND, and COAG to accommodate subtle differences in technique and accessory performance.
- Patient plate continuity monitoring with audible alarm.
- Distinct audible tones for CUT/BLEND modes and COAG mode with associated MODE light.
- Fully regulated isolated output power.
- Meets or exceeds IEC 601-2-2, second edition.
- Non-electric pneumatic foot pedal to maximize safety.
- Choice of reusable or disposable patient plate.
- Reusable handpiece.
- Choice of reusable or disposable electrodes.
- Output power safety audible alarm with automatic power shut off.
- Class 1, type BF, protected for use with defibrillator.
- Membrane switching to maximize cleanliness and ease of use.



Controls

1. Main switch
2. Socket for foot pedal
3. Socket for active electrodes
4. Socket for patient plate
5. Warning light of patient plate alarm (red)
6. Coagulation light (blue)
7. Pure cut and blend light (yellow)
8. Mode control



9. Power control
 - ⚡ Classification I Type BF protected against defibrillator effects
 - F Floating output circuit
 - ⚠ Cautions, consult this manual for safety precautions
 - 🦶 Pedal connection
 - 🔌 Active handle connection
 - 👤 Patient plate connection

Specifications:

Meets or Exceeds IEC 601-1-1, IEC 601-1-2 and IEC 601-2-2 specifications.
Meets the requirements of Annex 1 of Directive 93/42/EEC relating to Medical Devices. UL and CUL approved.

Electrical

Input Voltage:95 - 135 OAC 60 Hz
or
190 - 250 VAC 50 Hz
Current:2.3 amps max. or 1.2 amps max.
Low Frequency Leakage:.....Less than 50 microamps
Fuses:.....2.5 amps, T type
.....or 1.6 amps, T type

Physical

Dimensions:305mm x 267mm x 115mm
Weight:7.250 Kg

Electrosurgical Output

RF Output Frequency:.....450 kHz

RF Output Power:

		Volts p-p Max (open circuit)	Duty Cycle	Crest Factor
CUT0-100 watts RMS*	830	—	1.4	
BLEND0-100 watts RMS*	1200	60%	2.0	
COAG0-80 watts RMS*	3800	10%	5.5	

NOTE: *stable to >800 ohms (calibration at 500 ohms)

RF Isolation.....Less than 100 milliamps at 200 ohms (per IEC 601-2-2)

Classification.....I-Type BF

Output Circuit.....Floating output. Protected against the effects of the defibrillator.

Working Mode.....Discontinuous maximum duty cycle: 10/30 sec.

CoolingConvection cooling without fan

Control.....Foot pedal operated (pneumatic) with audible signals and mode lights

Audible signals and lights for operation and alarm:

Main.....green light

Alarm, patient plate continuity.....low pitch intermittent audible
alarm - red light

Alarm, output powerhigher pitch intermittent
audible alarm

Cut and Blend modeslow pitch audible signal - yellow light

Coagulation mode.....high pitch audible signal - blue light

NOTE: specifications subject to change

EQUIPMENT REQUIRED

DESCRIPTION	FEATURES	EXAMPLE
Double tracking oscilloscope	Passband: DC - 100 MHz	Tektronix Mod. 2235A
Digital multimeter	Accuracy: DC = 1%, AC = 2%	Hewlett Packard Mod. E2373A
Thermoelement	Passband: 10 Hz - 5 MHz	YOKOYAWA Mod. 2016
Digital frequency meter	Passband: 10 HZ - 100 MHz	Hewlett Packard Mod. HP 5384A
Voltage probe X10	Passband: DC - 200 MHz	Tektronix Mod. P6063B
Voltage probe X1000	Passband: DC - 10 MHz	Tektronix Mod. P6015
Current probe	Passband: DC - 50 MHz	Tektronix Mod. A6302/AM 503

A) PRELIMINARY PROCEDURES

- Remove the F3 fuse from the MAIN BOARD (cod. 801278).
- Insert plug from pedal control into input port. 
- Insert neutral plate plug in its appropriate outlet. 
- Insert power supply cable in current outlet.

B) LOW VOLTAGE POWER SUPPLY (BOARD 801278)

1) TESTING POWER SUPPLY VOLTAGE + 5V AND +15V (TP1, TP 2)

- Set oscilloscope vertical sensitivity at 1V/div. (plugged in DC).
- Connect oscilloscope probe to test point TP1.
- Turn on electrosurgical knife and make sure voltage is between 4.7V and 5.3V.
- Insert oscilloscope plug in AC outlet with vertical sensitivity set at 0.1V/div. and make sure that ripple voltage is not over 10mVpp.
- Move oscilloscope probe on test point TP2 (Vert. 2V/div.; DC) and make sure voltage is between 14.3V and 15.7V.

C) MICROCONTROLLER (BOARD 801282)

1) SELF TEST WHEN EQUIPMENT IS TURNED ON

Turn Power Switch "ON"

- Power Switch illuminates
- CUT, BLEND and COAG LED's illuminate
- Display reads "888"
- After approximately 10 seconds, display changes to "- - -" then CUT and BLEND activation LED's and buzzer triggers for a second.
- Display changes to "O"
- BLEND and COAG mode LED's deactivate

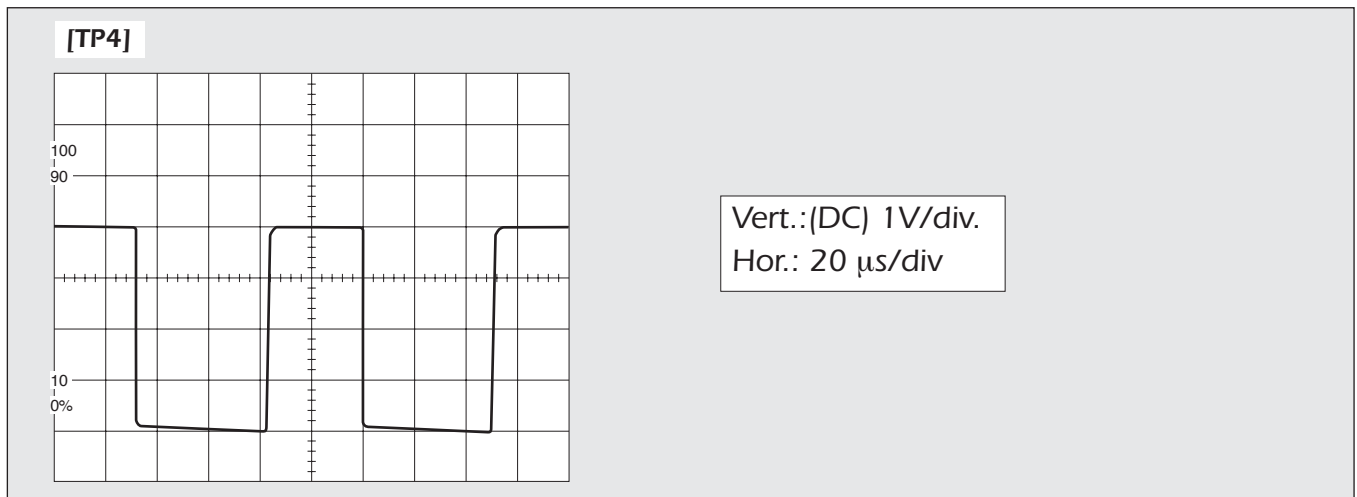
2) CALIBRATING VREF VOLTAGE

- Connect the voltmeter to test point TP3 (BOARD 801282).
- Turn trimmers R57 and R62 fully clockwise.
- Set trimmer R2 so that voltage is exactly at 4.94V.

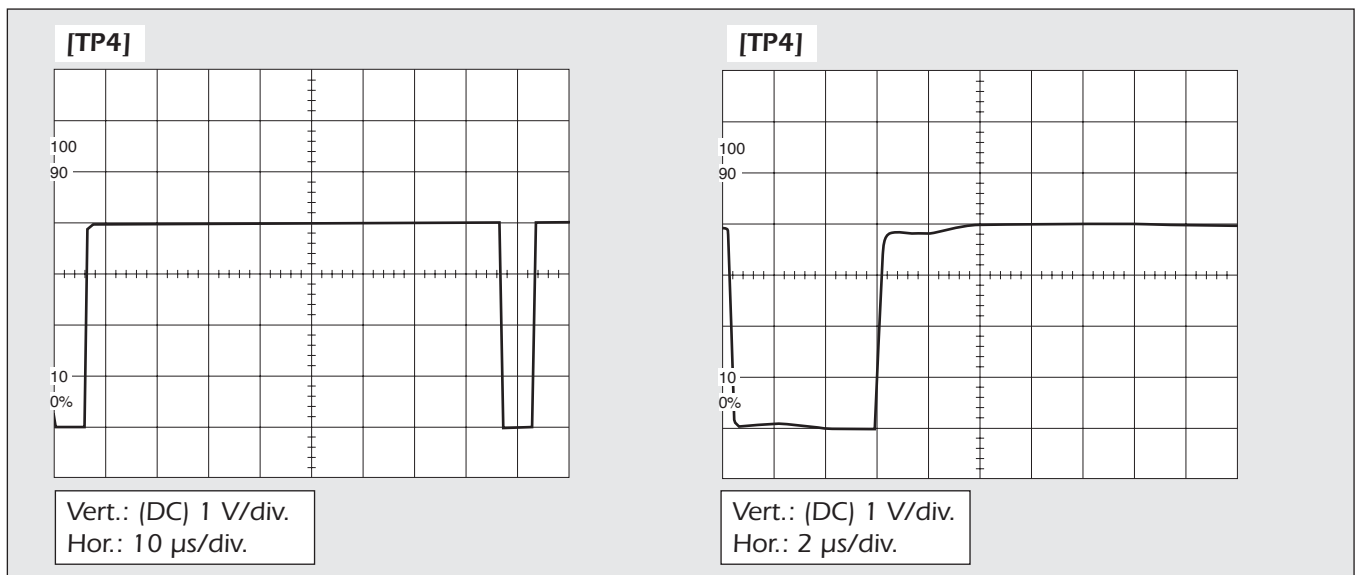
3) TESTING RF MODULATION SIGNAL

- Connect oscilloscope probe to test point TP4 (BOARD 801282).
- Set 10W as power index in the  function.

- c) With the electrosurgical knife turned on, test the modulation signal taking as reference the following wave form:



- e) Set 10W as power index in the  function. 
- f) With the electrosurgical knife turned on, test the modulation signal, taking as reference the following wave form:



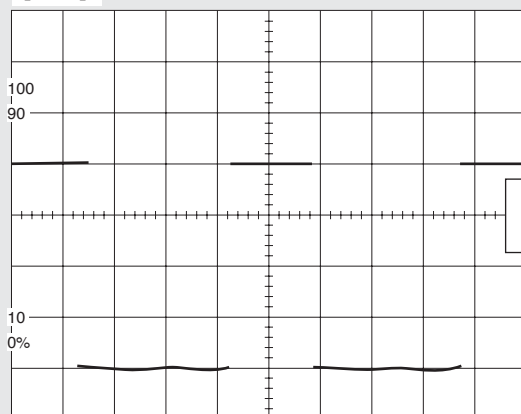
D) POWER SUPPLY (BOARD 801278)

1) CALIBRATING REFERENCE VOLTAGE VREFA

- Connect to TP5 to ground. (MOSFET drain Q6, Board 801278).
This prevents activation of the RF power generator, thus preventing possible damage to electronic circuits that have not been tested yet.
- Insert F3 fuse.
- Position trimmer R4 at center position (BOARD 801278).
- Turn on electrosurgical knife and wait for the display to show "0".
- Calibrate trimmer R6 (BOARD 801278) so that voltage at test point TP6 is exactly 8.0V.

2) TESTING PWM SIGNAL CONTROLLING THE POWER SUPPLIER

- Connect oscilloscope probe to test point TP7 (BOARD 801278).
- Set 100W as power index in the  function.
- Activate and make sure the following wave form appears:

[TP7]

Vert.: (DC) 2 V/div.
Hor.: 20 μ s/div.

Vert.:
(Gleich-)

3) CALIBRATING MINIMUM VOLTAGE

- Set 2W as power index in the function.
- Make sure that, when the electrosurgical knife is not turned on, voltage is lower than 3V in test point TP8.
- Press the activation pedal and bring voltage to 7V on TP8, by using trimmer R4.

4) CALIBRATING MAXIMUM VOLTAGE

- Set 80W as power index in the function.
- Press activation pedal and bring voltage to 134V on TP8, by using trimmer R6. Calibration of the two voltages, the maximum and minimum, is interactive; thus, it is necessary to repeat calibrations until the values that have been indicated are reached.
- Remove the jumper used to create electrical bonding on TP5 (MOSFET drain Q6, BOARD 801278).

E) RADIO FREQUENCY (BOARD 801278)

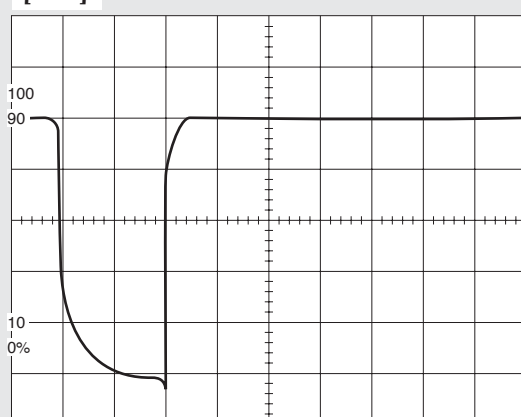
Remove F3 fuse

1) TESTING NEUTRAL CIRCUIT PLATE

- Select function.
- Disconnect the two end conductors in the plate and then insert a 1000 ohm resistor in series.
- Check that red LED is turned on, at the same time that the LED turns on, a repetitive acoustic signal will give warning of an error in the neutral plate.
- Once again, short the end conductors on the neutral plate.

2) TESTING WORKING FREQUENCY

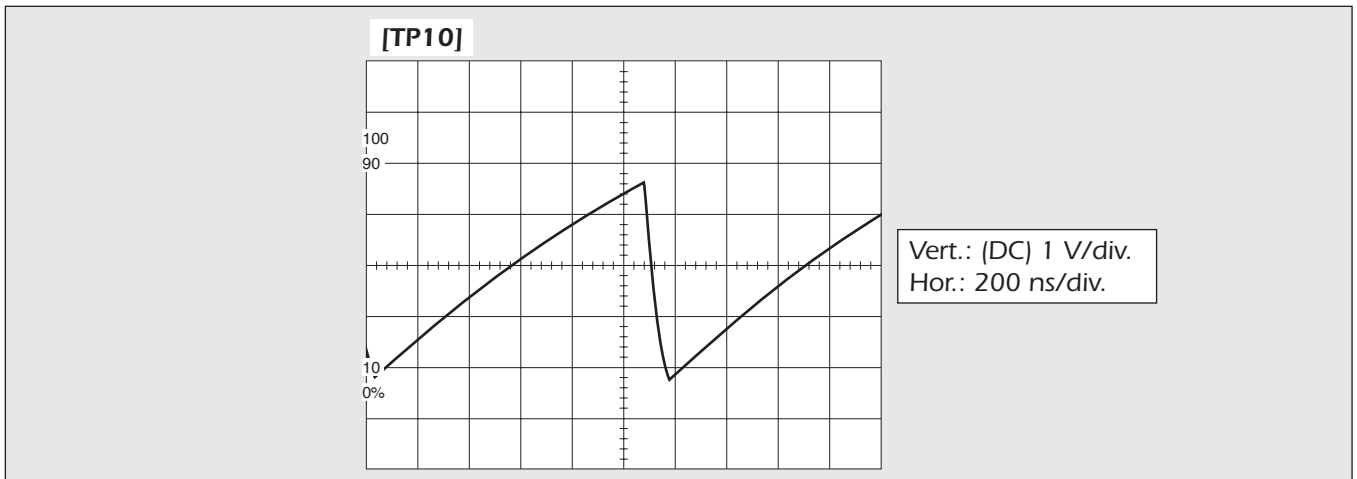
- Set 10W as power index in the function.
- Connect frequency meter to test point TP9.
- With the electrosurgical knife turned on, make sure that working frequency is 900 KHz (+ 10KHz).
- Calibrate the capacity trimmer C24 in such a way that the wave form is the following:

[TP9]

Vert.: (DC) 1 V/div.
Hor.: 50 ns/div.

3) CALIBRATING THE SLOPED SIGNAL (SAW-TOOTH SIGNAL)

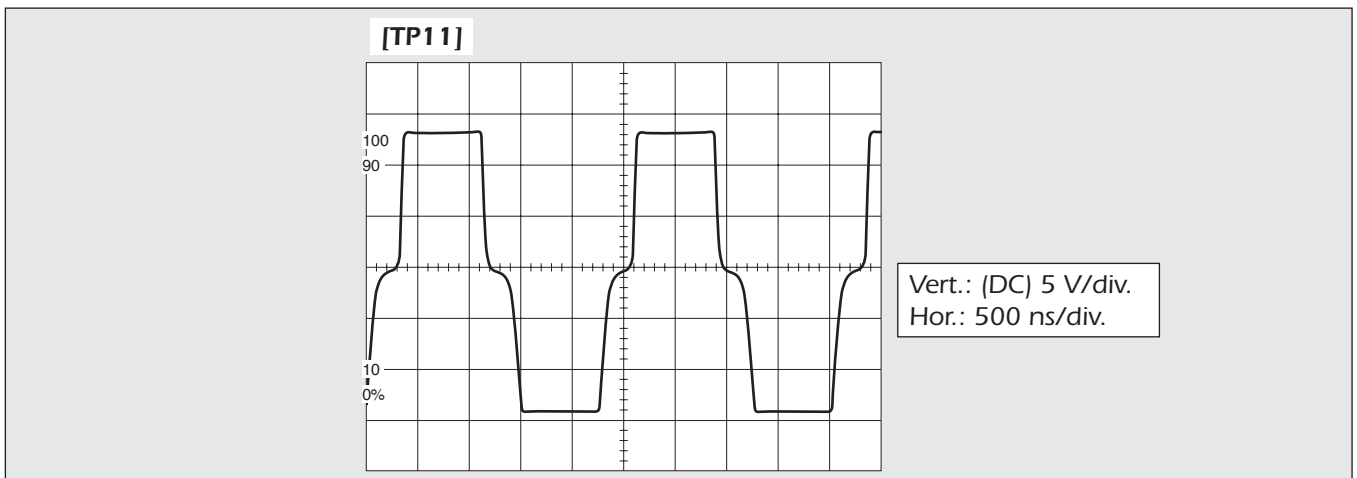
- a) Always with the CUT function activated, regulate the R53 trimmer in such a way that the wave form on test point TP10 will be the following:



- b) Turn the electrosurgical knife off.

4) CALIBRATING DEAD TIME

- a) Connect a load of 500 ohm to the  output hook up
b) Set 10W as power index in the  function.
c) Turn the R 61 trimmer fully counterclockwise.
d) Using both inlets of the oscilloscope, connect probe "1" to test point TP11, and probe "2" on test point TP12.
e) With the electrosurgical knife turned on, turn the R53 trimmer so that dead time (i.e., the time at which both signals are at zero level) is equal to 250ns.

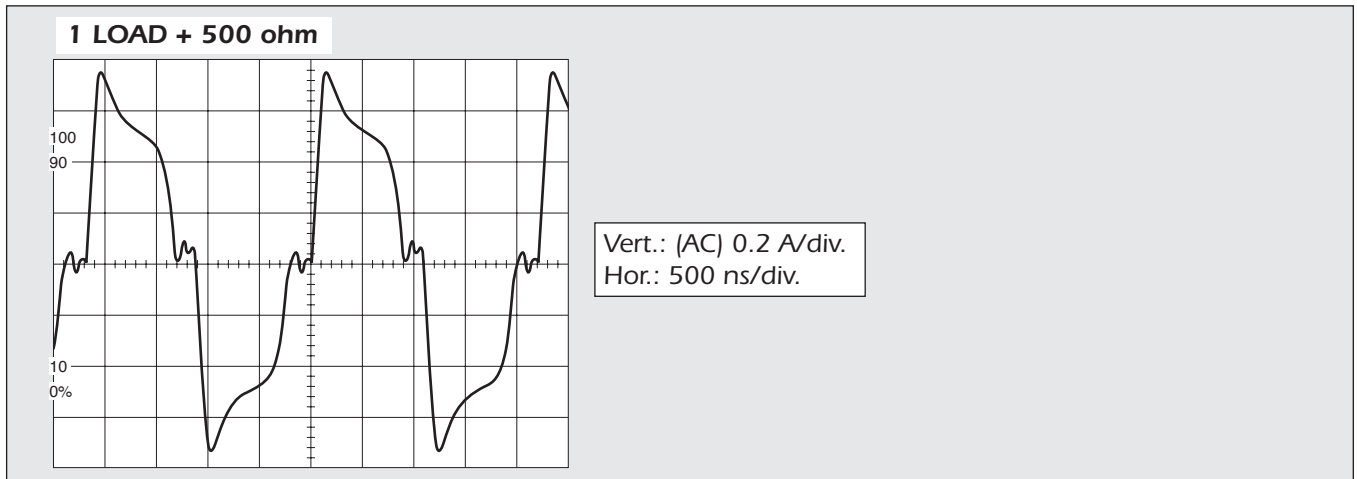


- g) Increase dead time by another 25ns by using trimmer R61.
h) Insert the F3 fuse.

5) TESTING THE RF POWER AMPLIFIER

- a) Set 100W as power index in the  function.
b) Insert the RF current probe in series in the output.

- c) activate the electrosurgical knife and make sure it has the following wave form:



6) CALIBRATING CURRENT LIMITER

- Bring output load to 200 ohm.
- Set 80W as power index in the $\rightarrow$ function.
- Calibrate the R76 trimmer in such a way that the current running through the output load is 580mA.

7) TESTING OUTPUT POWER

- Bring the output load to 500 ohm.
- With power set at 80W in $\rightarrow$, make sure that effective current in the load is equal to 400mA (+ 5%). If necessary, it is possible to adjust the maximum output power by intervening on the R6 trimmer. Calibrating the R6 trimmer is allowed only if the voltage that is supplying the final stage does not exceed 135V.
- Set 10W as power index in the $\leftarrow$ function
- Make sure that the effective current in the load is equal to 140mA (+ 5%). If necessary, it is possible to adjust the maximum output power by intervening on the R4 trimmer.
- Set 100W as power index in the $\leftarrow$ function
- Make sure that effective current in the load is equal to 450mA (+ 5%).

NOTE: This manual contains a chapter with drawings of the power output capacity curves for the electrosurgical knife; for a complete test of the device, refer to these curves, page 11.

8) CALIBRATING MAXIMUM LEVEL OF RF CURRENT LEAKAGE

- Bring load resistor to 1500 ohm.
- Set 100W as power index in the $\leftarrow$ function.
- Connect the oscilloscope probe to test point TP8.
- By calibrating the R115 trimmer, find the point in which the tension displayed in the oscilloscope decreases abruptly from 52V to 42V ($\pm 20\%$).

F) CALIBRATING RF PROTECTION SYSTEMS

1) PROTECTION FROM MALFUNCTION CAUSED BY MODULATION SIGNAL

- Bring load resistor to 1000 ohm.
- Set 40W as power index in the $\rightarrow$ function.
- With the electrosurgical knife activated, turn the R64 trimmer (BOARD 801282) slowly counterclockwise, until the audio signal becomes intermittent and, at the same time, the RF output signal stops.
- Now turn the R64 trimmer (BOARD 801282) clockwise 15 degrees
- Always keeping the power index at 40W, make sure that all functions are properly activated.
- In order to simulate a malfunction condition, short the R49 resistance (BOARD 801278) and activate in $\rightarrow$. The audio signal must warn of a malfunction condition.
- Remove jumper shorting the R49 resistance.

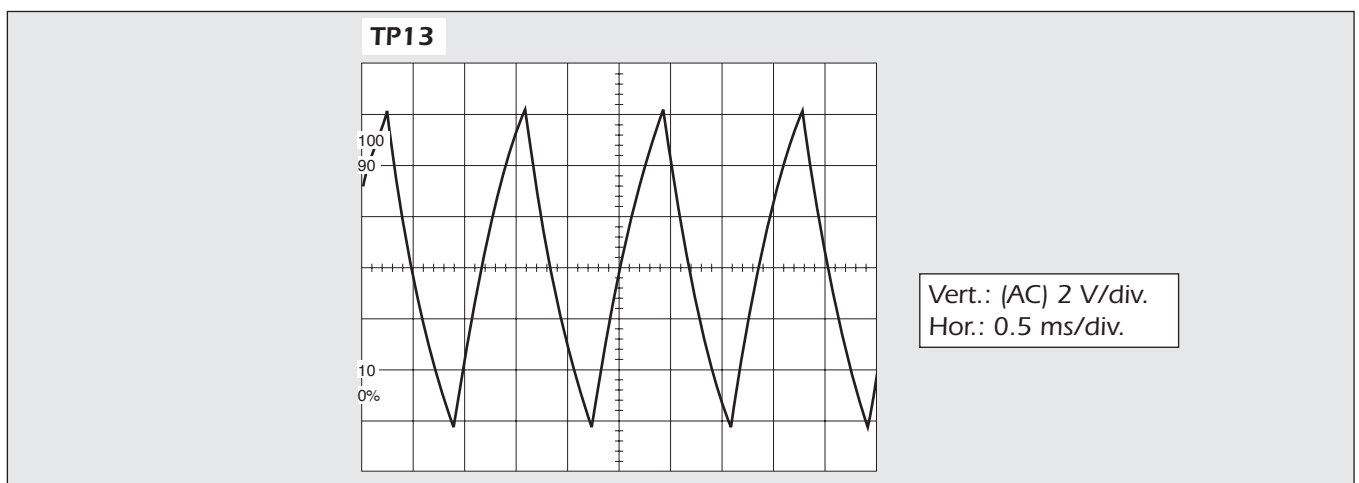
2) PROTECTION FROM MALFUNCTION CAUSED BY POWER SUPPLY

- Bring load resistor to 500 ohm.
- Set 40W as power index in the  function .
- With the electrosurgical knife activated, turn slowly the R59 trimmer counterclockwise (BOARD 801282) until the audio sound becomes intermittent.
- Now, turn the R59 trimmer (BOARD 801282) clockwise 15 degrees.
- Always keeping the power index at 40W, make sure that all functions are properly activated.

G) CHECKING THE AUDIO SIGNAL LEVEL

The audio level has no control. The frequency and amplitude of the signal require no calibrating and are generated directly by the microprocessor. To check, proceed as follows:

- Set 2W as power index of the  function.
- Connect the oscilloscope probe to test point TP13.
- Activate and make sure that the wave form is the following:

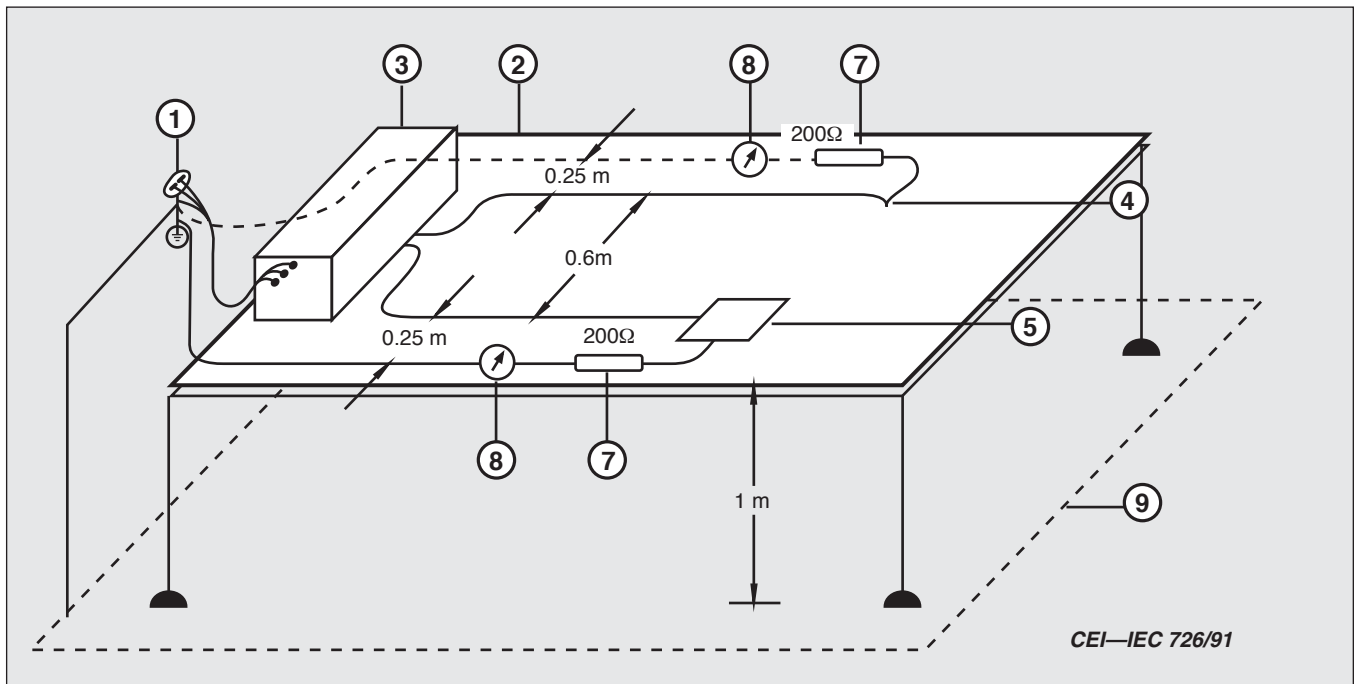


H) CHECKING RF CURRENT LEAKAGE

High frequency current leakage, for insulated equipment, must never exceed 150mA RMS, as required by the CEI EN 60601-2-2 REGULATIONS (IEC 601-2-2). The microprocessor for the electrosurgical knife LEEP 1000 has controls allowing it to keep well below such limit. To test the efficiency of these controls, it is indispensable to duplicate the most extreme test conditions, which in the case of the electrosurgical knife LEEP 1000 are obtained with the functions CUT, BLEND, FORCED COAG and SOFT COAG, at their highest power setting.

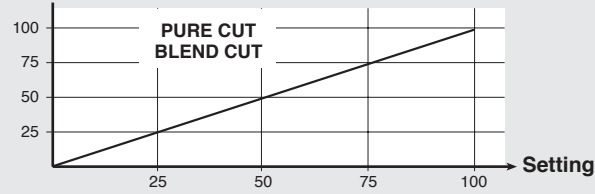
The average figures for leakage current are less than 100mA.

Measurements must be made as indicated by the CEI REGULATIONS EN 60601-2-2 (with insulated equipment).

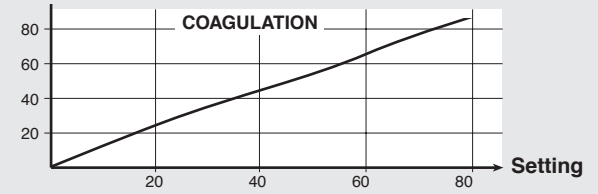


POWER OUTPUT CAPACITY CURVES

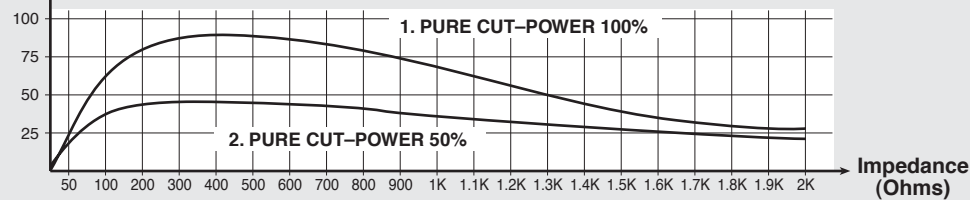
**POWER (500 Ohm)
(WW)**



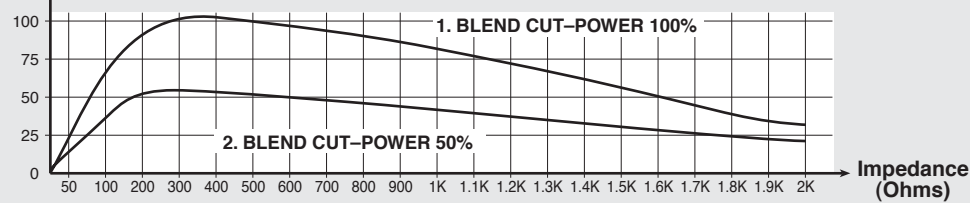
**POWER (500 Ohm)
(WW)**



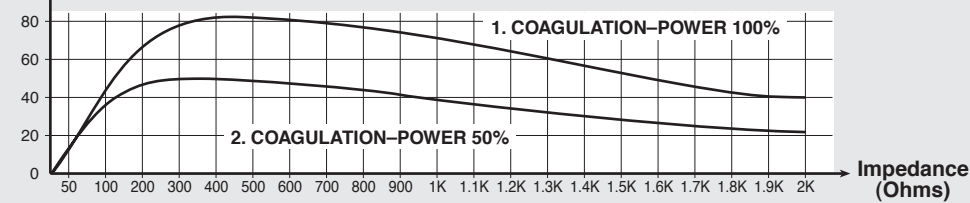
**POWER
(W)**



**POWER
(W)**



**POWER
(W)**

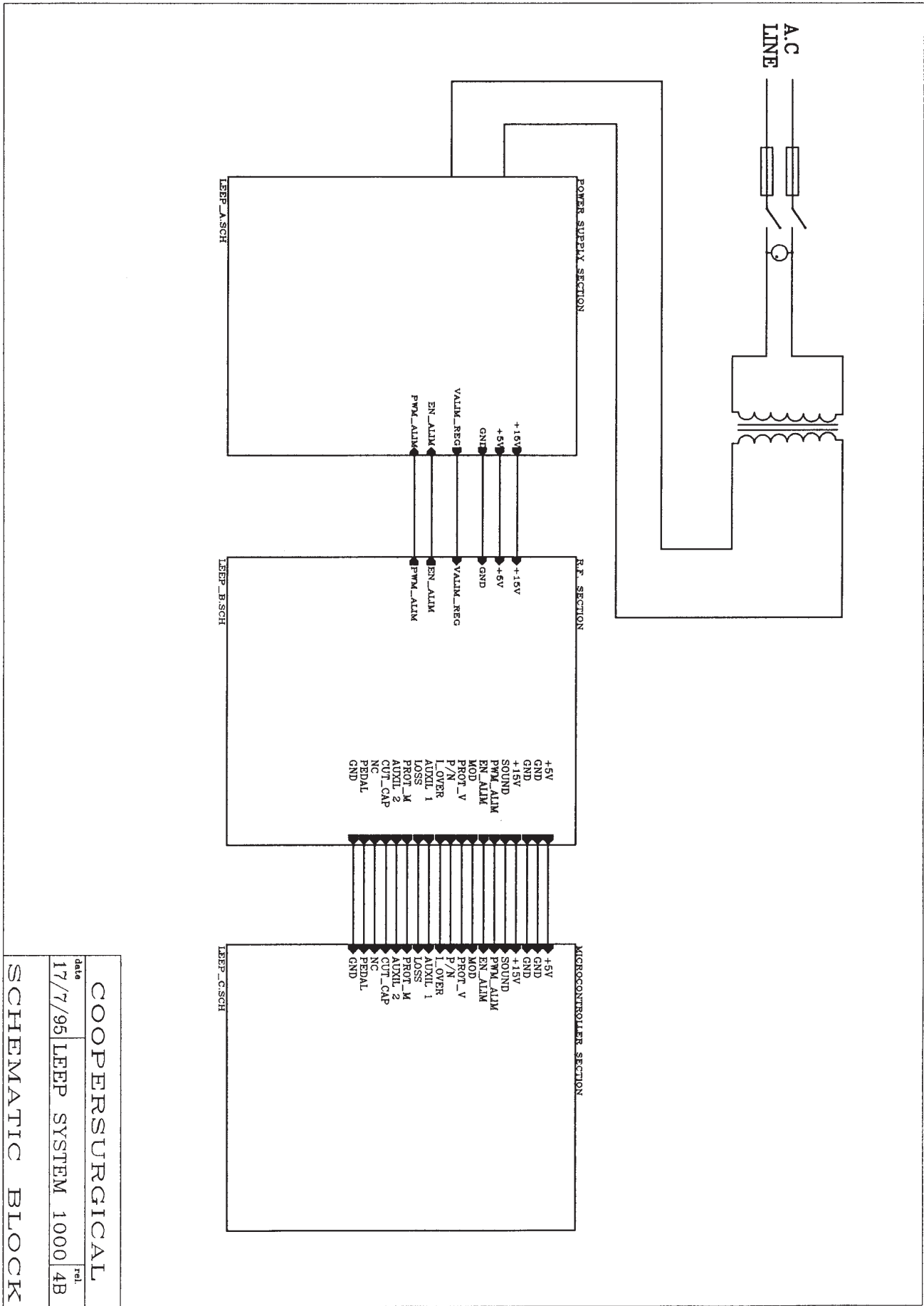


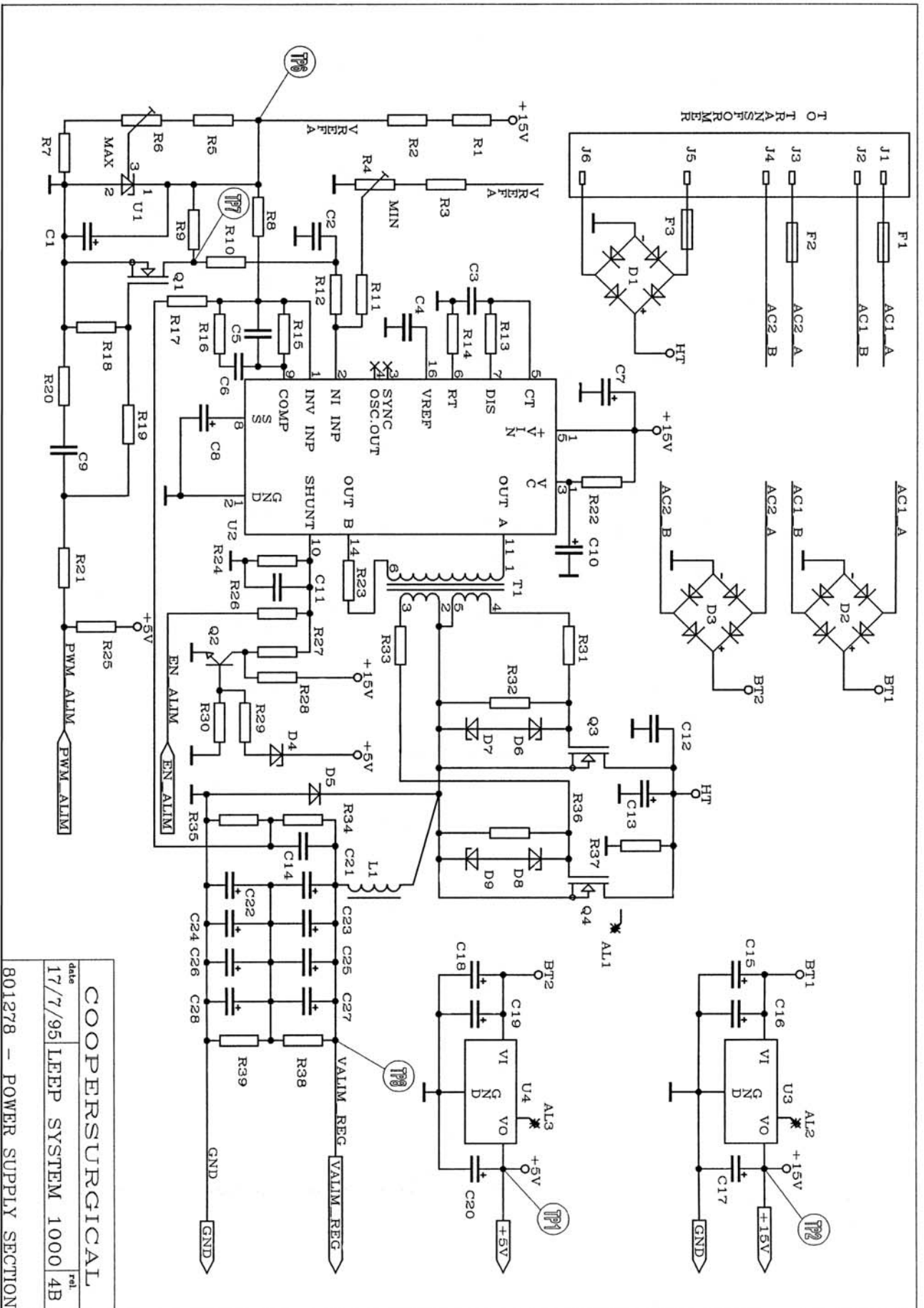
LEEP 1000		801253	Bill Of Materials	Revised May 3, 1994	Revision 4A
Reference	Part	Description		Code	
R1	330	Metal Film Resistor 1/4 W 5%		430171	
R2	330	Metal Film Resistor 1/4 W 5%		430171	
R3	3K9	Metal Film Resistor 1/4 W 5%		430350	
R4	10KT	Trimmer Cermet		403066	
R5	8K/1%	Precision Resistor 1% 0.5. W		430469	
R6	2KT	Trimmer Cermet		403092	
R7	3K/1%	Precision Resistor 1% 0.5. W		430459	
R8	150K	Metal Film Resistor 1/4 W 5%		430360	
R9	1K	Metal Film Resistor 1/4 W 5%		430170	
R10	100K	Metal Film Resistor 1/4 W 5%		430177	
R11	330K	Metal Film Resistor 1/4 W 5%		430173	
R12	10K	Metal Film Resistor 1/4 U 5%		430339	
R13	10	Metal Film Resistor 1/4 W 5%		430180	
R14	12K	Metal Film Resistor 1/4 W 5%		430351	
R15	220K	Metal Film Resistor 1/4 W 5%		430352	
R16	27K	Metal Film Resistor 1/4 W 5%		430340	
R17	10K	Metal Film Resistor 1/4 W 5%		430339	
R18	100K	Metal Film Resistor 1/4 W 5%		430177	
R19	1K	Metal Film Resistor 1/4 W 5%		430170	
R20	100	Metal Film Resistor 1/4 W 5%		430336	
R21	1K	Metal Film Resistor 1/4 W 5%		430170	
R22	10	Metal Film Resistor 1/4 W 5%		430180	
R23	10	Metal Film Resistor 1/4 W 5%		430180	
R24	22K	Metal Film Resistor 1/4 W 5%		430179	
R25	47K	Metal Film Resistor 1/4 W 5%		430218	
R26	10K	Metal Film Resistor 1/4 W 5%		430339	
R27	22K	Metal Film Resistor 1/4 W 5%		430179	
R28	47K	Metal Film Resistor 1/4 W 5%		430218	
R29	4K7	Metal Film Resistor 1/4 W 5%		430167	
R30	10K	Metal Film Resistor 1/4 W 5%		430339	
R31	56	Metal Film Resistor 1/4 W 5%		430444	
R32	1K	Metal Film Resistor 1/4 W 5%		430170	
R33	56	Metal Film Resistor 1/4 W 5%		430444	
R34	100K	Metal Film Resistor 1/4 W 5%		430177	
R35	5K6	Metal Film Resistor 1/4 W 5%		430344	
R36	1K	Metal Film Resistor 1/4 W 5%		430170	
R37	33K/6W/F	Wirewound Resistor 33 K 6 W		430463	
R38	10K/1W	Metal Film Resistor 1 W		430320	
R39	10K/1W	Metal Film Resistor 1 W		430320	
R40	10K	Metal Film Resistor 1/4 W 5%		430339	
R41	22K	Metal Film Resistor 1/4 W 5%		430179	
R42	12K	Metal Film Resistor 1/4 W 5%		430351	
R43	10	Metal Film Resistor 1/4 W 5X		430180	
R44	1K8	Metal Film Resistor 1/4 W 5%		430198	
R45	100	Metal Film Resistor 1/4 W 5%		430336	
R46	1K8	Metal Film Resistor 1/4 W 5%		430198	
R47	220	Metal Film Resistor 1/4 W 5%		430358	
R48	1K8	Metal Film Resistor 1/4 W 5%		430198	
R49	4K7	Metal Film Resistor 1/4 W 5%		430167	
R50	220	Metal Film Resistor 1/4 W 5%		430358	
R51	10K	Metal Film Resistor 1/4 W 5%		430339	
R52	100	Metal Film Resistor 1/4 W 5%		430336	
R53	10KT	Trimmer Cermet		403066	
R54	12K	Metal Film Resistor 1/4 W 5%		430351	
R55	47	Metal Film Resistor 1/4 W 5%		430334	
R56	39K	Metal Film Resistor 1/4 W 5%		430217	
R57	27K	Metal Film Resistor 1/4 W 5%		430340	
R58	1K	Metal Film Resistor 1/4 W 5%		430170	
R59	1K8	Metal Film Resistor 1/4 W 5%		430198	
R60	100K	Metal Film Resistor 1/4 W 5%		430177	
R61	100KT	Trimmer Cermet		403084	
R62	560	Metal Film Resistor 1/4 W 5%		430178	
R63	27K	Metal Film Resistor 1/4 W 5%		430340	
R64	56K	Metal Film Resistor 1/4 W 5%		430176	
R65	560	Metal Film Resistor 1/4 W 5%		430178	
R66	27K	Metal Film Resistor 1/4 W 5%		430340	
R67	27K	Metal Film Resistor 1/4 W 5%		430340	
R68	4K7	Metal Film Resistor 1/4 W 5%		430167	
R69	22	Metal Film Resistor 1/4 W 5%		430447	
R70	22	Metal Film Resistor 1/4 W 5%		430447	
R71	10	Metal Film Resistor 1/4 W 5%		430180	
R72	100K	Metal Film Resistor 1/4 W 5%		430177	
R73	1K	Metal Film Resistor 1/4 W 5%		430170	
R74	10K	Metal Film Resistor 1/4 W 5%		430339	
R75	2K2	Metal Film Resistor 1/4 W 5%		430343	
R76	5KT	Trimmer Cermet		403052	
R77	1K	Metal Film Resistor 1/4 W 5%		430170	
R78	4K7	Metal Film Resistor 1/4 W 5%		430167	
R79	47K	Metal Film Resistor 1/4 W 5%		430218	
R80	10K	Metal Film Resistor 1/4 W 5%		430339	

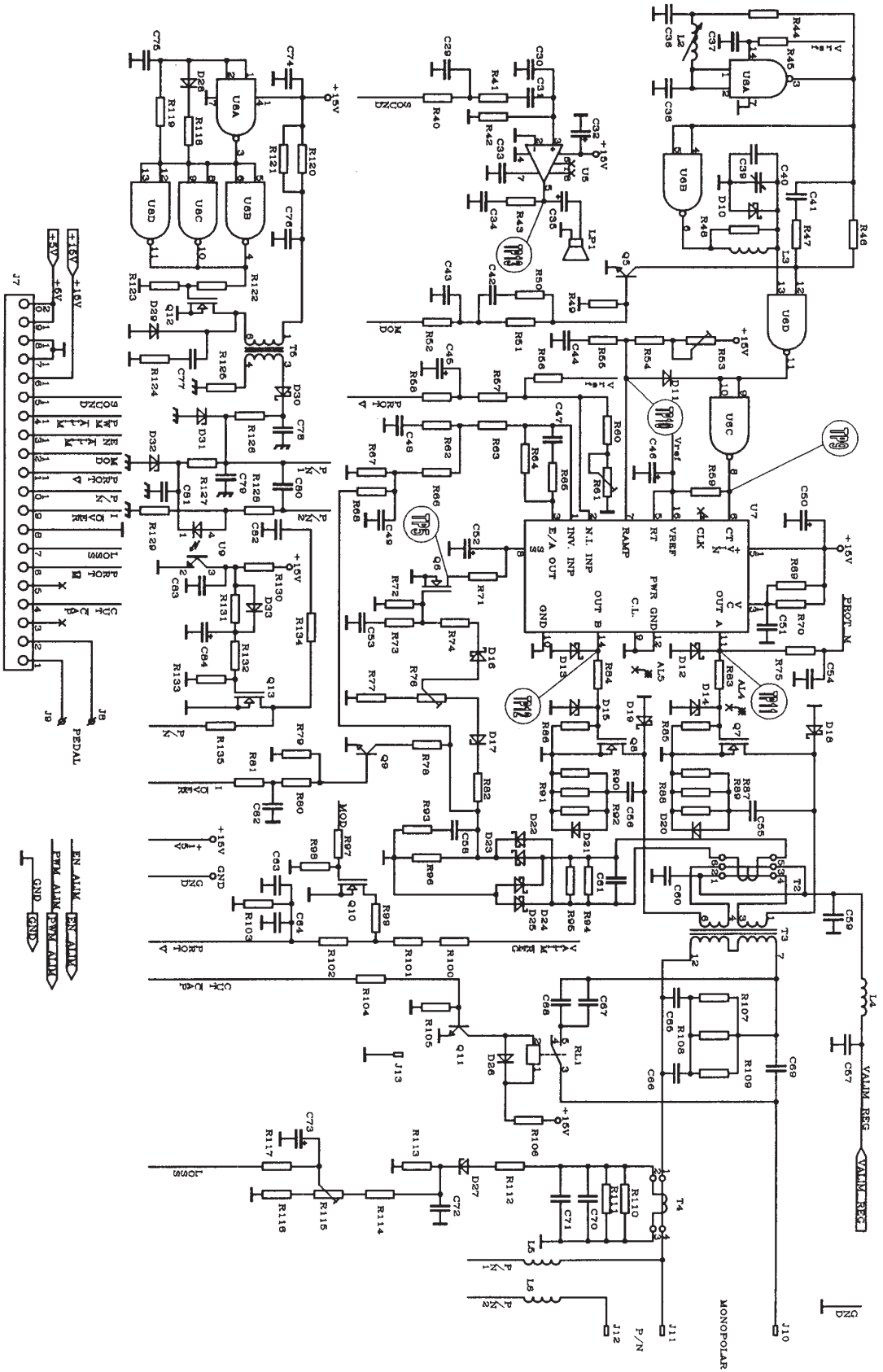
Reference	Part	Description	Code
R81	10K	Metal Film Resistor 1/4 W 5%	430339
R82	1K	Metal Film Resistor 1/4 W 5%	430170
R83	12	Metal Film Resistor 1/4 W 5%	430415
R84	12	Metal Film Resistor 1/4 W 5%	430415
R85	1K	Metal Film Resistor 1/4 W 5%	430170
R86	1K	Metal Film Resistor 1/4 W 5%	430170
R87	1K/3W/A	Non Inductive Resistor 3 W	430450
R88	1K/3W/A	Non Inductive Resistor 3 W	430450
R89	1K/3W/A	Non Inductive Resistor 3 W	430450
R90	1K/3W/A	Non Inductive Resistor 3 W	430450
R91	1K/3W/A	Non Inductive Resistor 3 W	430450
R92	1K/3W/A	Non Inductive Resistor 3 W	430450
R93	100	Metal Film Resistor 1/4 W 5%	430336
R94	470	Metal Film Resistor 1/4 W 5%	430169
R95	470	Metal Film Resistor 1/4 W 5%	430169
R96	2K2	Metal Film Resistor 1/4 W 5%	430343
R97	470	Metal Film Resistor 1/4 W 5%	430169
R98	22K	Metal Film Resistor 1/4 W 5%	430179
R99	2K/1%	Precision Resistor 1% 0.5 W	430461
R100	11K/1%	Precision Resistor 1% 0.5 W	430467
R101	11K/1%	Precision Resistor 1% 0.5 W	430467
R102	11K/1%	Precision Resistor 1% 0.5 W	430467
R103	33K2/1%	Precision Resistor 1% 0.5 W	430472
R104	4K7	Metal Film Resistor 1/4 W 5%	430167
R105	10K	Metal Film Resistor 1/4 W 5%	430339
R106	47	Metal Film Resistor 1/4 W 5%	430334
R107	10K/5W	Non Inductive Resistor 5 W	430471
R108	10K/5W	Non Inductive Resistor 5 W	430471
R109	10K/5W	Non Inductive Resistor 5 W	430471
R110	2K2	Metal Film Resistor 1/4 W 5%	430343
R111	2K2	Metal Film Resistor 1/4 W 5%	430343
R112	12	Metal Film Resistor 1/4 W 5%	430415
R113	4K7	Metal Film Resistor 1/4 W 5%	430167
R114	10K	Metal Film Resistor 1/4 W 5%	430339
R115	10KT	Trimmer Cermet	403066
R116	1K	Metal Film Resistor 1/4 W 5%	430170
R117	100	Metal Film Resistor 1/4 W 5%	430336
R118	22K	Metal Film Resistor 1/4 W 5%	430179
R119	33K	Metal Film Resistor 1/4 W 5%	430172
R120	220	Metal Film Resistor 1/4 W 5%	430358
R121	220	Metal Film Resistor 1/4 W 5%	430358
R122	100	Metal Film Resistor 1/4 W 5%	430336
R123	680	Metal Film Resistor 1/4 W 5%	430194
R124	180	Metal Film Resistor 1/4 W 5%	430342
R125	10	Metal Film Resistor 1/4 W 5%	430180
R126	100	Metal Film Resistor 1/4 W 5%	430336
R127	1KS	Metal Film Resistor 1/4 W 5%	430199
R128	560	Metal Film Resistor 1/4 W 5%	430178
R129	820	Metal Film Resistor 1/4 W 5%	430200
R130	18K	Metal Film Resistor 1/4 W 5%	430346
R131	100K	Metal Film Resistor 1/4 W 5%	430177
R132	2K2	Metal Film Resistor 1/4 W 5%	430343
R133	1M	Metal Film Resistor 1/4 W 5%	430331
R134	NU	NOT USED	-----
R135	100	Metal Film Resistor 1/4 W 5%	430336
C1	10u/T/35V	Tantalum Electrolytic Capacitor	400134
C2	100n	Ceramic Capacitor	400139
C3	1n/P	Metallized Polyester Capacitor	400278
C4	100n	Ceramic Capacitor	400139
C5	100p/NPO	NPO Ceramic Capacitor	400260
C6	220P/C	Ceramic Capacitor	400240
C7	1 Ou/T/35V	Tantalum Electrolytic Capacitor	400134
C8	1 u/T/35V	Tantalum Electrolytic Capacitor	400173
C9	22p/C	Ceramic Capacitor	400290
C10	10u/T/35V	Tantalum Electrolytic Capacitor	400134
C11	100n	Ceramic Capacitor	400139
C12	10n/C/1KV	Ceramic Capacitor	400133
C13	470u/E/250V	Vertical Electrolytic Capacitor	400304
C14	680p/C/500V	Ceramic Capacitor	400292
C15	1000u/E/35V	Electrolytic Capacitor	400256
C16	10u/T/35V	Tantalum Electrolytic Capacitor	400134
C17	10u/T/35V	Tantalum Electrolytic Capacitor	400134
C18	1000u/E/35V	Electrolytic Capacitor	400256
C19	1 Ou/T/35V	Tantalum Electrolytic Capacitor	400134
C20	1 Ou/T/35V	Tantalum Electrolytic Capacitor	400134
C21	22u/E/100V/V	Vertical Electrolytic Capacitor SMPS	400286
C22	22u/E/100V/V	Vertical Electrolytic Capacitor SMPS	400286
C23	22u/E/100V/V	Vertical Electrolytic Capacitor SMPS	400286

Reference	Part	Description	Code
C24	22u/E/100V/V	Vertical Electrolytic Capacitor SMPS	400286
C25	22u/E/100V/V	Vertical Electrolytic Capacitor SMPS	400286
C26	22u/E/100V/V	Vertical Electrolytic Capacitor SMPS	400286
C27	22u/E/100V/V	Vertical Electrolytic Capacitor SMPS	400286
C28	22u/E/100V/V	Vertical Electrolytic Capacitor SMPS	400286
C29	100n	Ceramic Capacitor	400139
C30	100p/NPO	NPO Ceramic Capacitor	400260
C31	100n	Ceramic Capacitor	400139
C32	1 Ou/T/35V	Tantalum Electrolytic Capacitor	400134
C33	10n	Ceramic Capacitor	400251
C34	100n	Ceramic Capacitor	400139
C35	22u/E/35V	Vertical Electrolytic Capacitor	400237
C36	1n/P	Metallized Polyester Capacitor	400278
C37	100n	Ceramic Capacitor	400139
C38	2nZ/P	Metallized Polyester Capacitor	400277
C39	12p/C	Ceramic Capacitor	400269
C40	7_100p	Trimmer Capacitor 7/100p	400289
C41	220P/C	Ceramic Capacitor	400240
C42	47p/NPo	NPO Ceramic Capacitor	400255
C43	100p/NPO	NPO Ceramic Capacitor	400260
C44	220p/PP/400V/A	Axial Polypropylene Capacitor	400280
C45	10u/T/35V	Tantalum Electrolytic Capacitor	400134
C46	10u/T/35V	Tantalum Electrolytic Capacitor	400134
C47	100n	Ceramic Capacitor	400139
CU	100n	Ceramic Capacitor	400139
C49	10n	Ceramic Capacitor	400251
C50	10u/T/35V	Tantalum Electrolytic Capacitor	400134
C51	100n	Ceramic Capacitor	400139
C52	1 u/T/35V	Tantalum Electrolytic Capacitor	400173
C53	100n	Ceramic Capacitor	400139
C54	220n/P	Metallized Film Capacitor	400271
C55	2n2/C/3KV	High Voltage Ceramic Capacitor	400223
C56	2n2/C/3KV	High Voltage Ceramic Capacitor	400223
C57	10n/C/1KV	Ceramic Capacitor	400133
C58	10n	Ceramic Capacitor	400251
C59	10n/C/1KV	Ceramic Capacitor	400133
C60	220n/PP/400V	Polypropylene Capacitor (SIEMENS)	400297
C61	1n/P	Metallized Polyester Capacitor	400278
C62	100n	Ceramic Capacitor	400139
C63	220n/P	Metallized Film Capacitor	400271
C64	220n/P	Metallized Film Capacitor	400271
C65	47p/C/6KV	High Voltage Ceramic Capacitor	400301
C66	47p/C/6KV	High Voltage Ceramic Capacitor	400301
C67	1n/PP/2KV	Polypropylene Capacitor (Philips)	400299
C68	1n/PP/2KV	Polypropylene Capacitor (Philips)	400299
C69	1n/PP/2KV	Polypropylene Capacitor (Philips)	400299
C70	680p/C/500V	Ceramic Capacitor	400292
C71	680p/C/500V	Ceramic Capacitor	400292
C72	10n	Ceramic Capacitor	400251
C73	1 u/T/35V	Tantalum Electrolytic Capacitor	400173
C74	100n	Ceramic Capacitor	400139
C75	1n/P	Metallized Polyester Capacitor	400278
C76	100n	Ceramic Capacitor	400139
C77	2n2/C	Ceramic Capacitor	400261
C78	220n/P	Metallized Film Capacitor	400271
C79	100n	Ceramic Capacitor	400139
C80	100n	Ceramic Capacitor	400139
C81	100n	Ceramic Capacitor	400139
C82	100n	Ceramic Capacitor	400139
C83	100n	Ceramic Capacitor	400139
C84	10u/T/35V	Tantalum Electrolytic Capacitor	400134
D1	KBL06	Bridge Rectifier 4A/600V	420069
D2	WL04	Bridge Rectifier 1A	420013
D3	WL04	Bridge Rectifier 1A	420013
D4	2V7/O SW	Zener diode 2V7-1/2W	420067
D5	BYT03/400	Ultra Fast Diode 3A/400V	420071
D6	12V/1W	Zener diode 12V/1W	420019
D7	12V/1W	Zener diode 12V/1W	420019
D8	12V/1W	Zener diode 12V/1W	420019
D9	12V/1W	Zener diode 12V/ 1W	420019
D10	BAT83	Schottky Diode	420070
D11	1N4148	Diode	420010
D12	11DQ06	Schottky Diode	420054
D13	11DQ06	Schottky Diode	420054
D14	12V/1W	Zener diode 12V/1W	420019
D15	12V/1W	Zener diode 12V/1W	420019
D16	BAT83	Schottky Diode	420070
D17	10V/O.5W	Zener diode 10V-1/2W	420068
D18	1 50V/5W	Zener Diode (x2)	420091

LEEP 1000	801253	Bill Of Materials	Revised May 3, 1994	Revision 4A
Reference	Part	Description	Code	
D19	150V/5W	Zener Diode 150V/5W(x2)	420091	
D20	BYT03/400	Ultra Fast Diode 3A/400V	420071	
D21	BYT03/400	Ultra Fast Diode 3A/400V	420071	
D22	11DQ06	Schottky Diode	420054	
D23	11DQ06	Schottky Diode	420054	
D24	11DQ06	Schottky Diode	420054	
D25	11DQ06	Schottky Diode	420054	
D26	1N4007	Diode	420001	
D27	11DQ06	Schottky Diode	420054	
D28	1N4148	Diode	420010	
D29	1N4148	Diode	420010	
D30	11DQ06	Schottky Diode	420054	
D31	10V/0.5W	Zener diode 10V-1/2W	420068	
D32	2V7/0.5W	Zener diode 2V7-1/2W	420067	
D33	1N4148	Diode	420010	
Q1	VN10KM	N Channel Mosfet	427054	
Q2	BC237B	NPN Transistor	427057	
Q3	IRF740	N Channel Mosfet	427076	
Q4	IRF740	N Channel Mosfet	427076	
Q5	BC237B	NPN Transistor	427057	
Q6	VN10KM	N Channel Mosfet	427054	
Q7	IRF740(IR)	N Channel Mosfet (IR)	427086	
Q8	IRF740(IR)	N Channel Mosfet (IR)	427086	
Q9	BC237B	NPN Transistor	427057	
Q10	VN10KM	N Channel Mosfet	427054	
Q11	BC237B	NPN Transistor	427057	
Q12	VN10KM	N Channel Mosfet	427054	
Q13	VN10KM	N Channel Mosfet	427054	
L1	520uH	Toroidal Inductor	713623	
L2	703191	Variable Inductor 33uH	703191	
L3	150uH/175mA	Inductor	422008	
L4	713621	Ind AMIDON T94-2	713621	
L5	1mH/100mA	Inductor	422005	
L6	1mH/100mA	Inductor	422005	
U1	LM431	Voltage Regulator	482078	
U2	UC3525	PWM UC3525	482060	
U3	LM78S15	Voltage Regulator +15 V	482086	
U4	LM7805	Voltage Regulator LM7805	482003	
U5	LM386	BF LM386	482056	
U6	74HCT00	HCT 7400	482049	
U7	UC3825	PWM UC3825	482064	
U8	4093	CMOS CD4093	482025	
U9	CNY65	Optocoupler CNY 65	482064	
RL1	V23061	Relay Siemens	404044	
T1	421026	Pulse Transformer (2 secondaries)	421026	
T2	713337	H F Current Transformer	713337	
T3	801255	Output Transformer LEEP SYSTEM 1000	801255	
T4	713708	H F Current Transformer	713708	
T5	713744	Transformer 10KV (SIRIO)	713744	
F1	1A/T	Fuse 1 A T 5x20 mm	433009	
F2	1A/T	Fuse 1 A T 5x20 mm	433009	
F3	2A/T	Fuse 2 A T 5x20 mm	433001	
J1	399028	Connector Faston C.S.	399028	
J2	399028	Connector Faston C.S.	399028	
J3	399028	Connector Faston C.S.	399028	
J4	399028	Connector Faston C.S.	399028	
J5	399028	Connector Faston C.S.	399028	
J6	399028	Connector Faston C.S.	399028	
J7	STRIP_20P	Strip 20 pin	446015/20	
J8	399028	Connector Faston C.S.	399028	
J9	399028	Connector Faston C.S.	399028	
J10	399028	Connector Faston C.S.	399028	
J11	399028	Connector Faston C.S.	399028	
J12	399028	Connector Faston C.S.	399028	
J13	399028	Connector Faston C.S.	399028	
LP1	449125	Loudspeaker 8 ohm 0 25 W	449125	
AL1	ML73_2	Aluminium Heat Sink	713249	
AL2	ML73_2	Aluminium Heat Sink	713249	
AL3	ML73_2	Aluminium Heat Sink	713249	
AL4	ML73_2	Aluminium Heat Sink	713249	
AL5	ML73_2	Aluminium Heat Sink	713249	

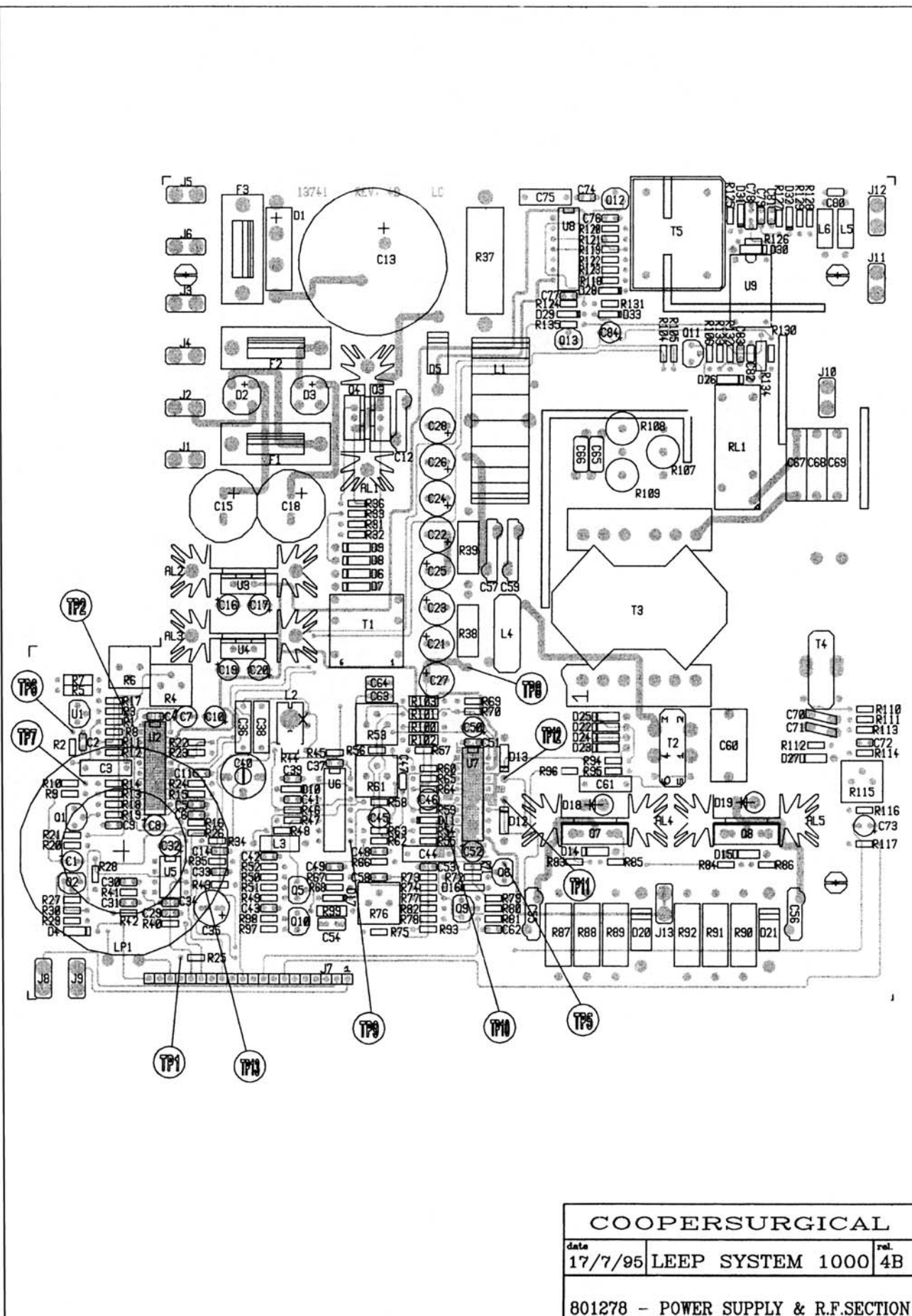






c2

COOPERSURGICAL	
date	17/7/95
LEEP SYSTEM 1000 4B	
B01278 - R.F. SECTION	



c3

		Description	Code
R1	560	Metal Film Resistor 1/4 W 5%	430178
R2	5KT	Trimmer Cermet	403052
R3	1K/1%	Precision Resistor 1% 0.5 W	430458
R4	3K/1%	Precision Resistor 1% 0.5 W	430459
R5	1K	Metal Film Resistor 1/4 W 5%	430170
R6	100	Metal Film Resistor 1/4 W 5%	430336
R7	1K	Metal Film Resistor 1/4 W 5%	430170
R8	10K	Metal Film Resistor 1/4 W 5%	430339
R9	1K8	Metal Film Resistor 1/4 W 5%	430198
R10	22K	Metal Film Resistor 1/4 W 5%	430179
R11	10K	Metal Film Resistor 1/4 W 5%	430339
R12	10K	Metal Film Resistor 1/4 W 5%	430339
R13	10K	Metal Film Resistor 1/4 W 5%	430339
R14	100	Metal Film Resistor 1/4 W 5%	430336
R15	100	Metal Film Resistor 1/4 W 5%	430336
R16	100	Metal Film Resistor 1/4 W 5%	430336
R17	10K	Metal Film Resistor 1/4 W 5%	430336
R18	10K	Metal Film Resistor 1/4 W 5%	430339
R19	10K	Metal Film Resistor 1/4 W 5%	430339
R20	1K	Metal Film Resistor 1/4 W 5%	430170
R21	1K	Metal Film Resistor 1/4 W 5%	430170
R22	1K	Metal Film Resistor 1/4 W 5%	430336
R23	100	Metal Film Resistor 1/4 W 5%	430336
R24	1K	Metal Film Resistor 1/4 W 5%	430170
R25	10K	Metal Film Resistor 1/4 W 5%	430339
R26	10K	Metal Film Resistor 1/4 W 5%	430339
R27	10K	Metal Film Resistor 1/4 W 5%	430170
R28	1K	Metal Film Resistor 1/4 W 5%	430170
R29	1K	Metal Film Resistor 1/4 W 5%	430170
R30	100	Metal Film Resistor 1/4 W 5%	430336
R31	100	Metal Film Resistor 1/4 W 5%	430336
R32	80/1%	Precision Resistor 1% 0.5 W	430460
R33	3K/1%	Precision Resistor 1% 0.5 W	430459
R34	80/1%	Precision Resistor 1% 0.5 W	430460
R35	3K/1%	Precision Resistor 1% 0.5 W	430459
R36	3K/1%	Precision Resistor 1% 0.5 W	430459
R37	22K	Metal Film Resistor 1/4 W 5%	430179
R38	22K	Metal Film Resistor 1/4 W 5%	430179
R39	560K	Metal Film Resistor 1/4 W 5%	430178
R40	10K	Metal Film Resistor 1/4 W 5%	430339
R41	22K	Metal Film Resistor 1/4 W 5%	430179
R42	10K	Metal Film Resistor 1/4 W 5%	430339
R43	1K	Metal Film Resistor 1/4 W 5%	430170
R44	10K	Metal Film Resistor 1/4 W 5%	430339
R45	10K	Metal Film Resistor 1/4 W 5%	430339
R46	10K	Metal Film Resistor 1/4 W 5%	430339
R47	10K	Metal Film Resistor 1/4 W 5%	430339
R48	10K	Metal Film Resistor 1/4 W 5%	430339
R49	10K	Metal Film Resistor 1/4 W 5%	430339
R50	10K	Metal Film Resistor 1/4 W 5%	430339
R51	560	Metal Film Resistor 1/4 W 5%	430178
R52	470	Metal Film Resistor 1/4 W 5%	430169
R53	470	Metal Film Resistor 1/4 W 5%	430169
R54	10K	Metal Film Resistor 1/4 W 5%	430339
R55	10K	Metal Film Resistor 1/4 W 5%	430339
R56	10K	Metal Film Resistor 1/4 W 5%	430339
R57	100	Metal Film Resistor 1/4 W 5%	403458
R58	1K/1%	Precision Resistor 1% 0.5 W	430170
R59	20K	Trimmer Cermet	403050
R60	1K	Metal Film Resistor 1/4 W 5%	430170
R61	10K	Metal Film Resistor 1/4 W 5%	430339
R62	NU	NOT USED	-----
R63	1K/1%	Precision Resistor 1% 0.5 W	430458
R64	50K	Trimmer Cermet	430088
R65	1K/1%	Precision Resistor 1% 0.5 W	430458
R66	1K	Metal Film Resistor 1/4 W 5%	430170
R67	NU	NOT USED	-----
R68	NU	NOT USED	-----
R69	100	Metal Film Resistor 1/4 W 5%	400336
C1	10n	Ceramic Capacitor	400251
C2	47p/NPO	NPO Ceramic Capacitor	400255
C4	100n	Ceramic Capacitor	400139
C5	100n	Ceramic Capacitor	400139
C6	100n	Ceramic Capacitor	400139
C7	100n	Ceramic Capacitor	400139
C8	100n	Ceramic Capacitor	400139
C9	100n	Ceramic Capacitor	400139
C10	10u/T/35V	Tantalum Electrolytic Capacitor	400134
C11	22p1C	Ceramic Capacitor	400290
C12	22p/C	Ceramic Capacitor	400290

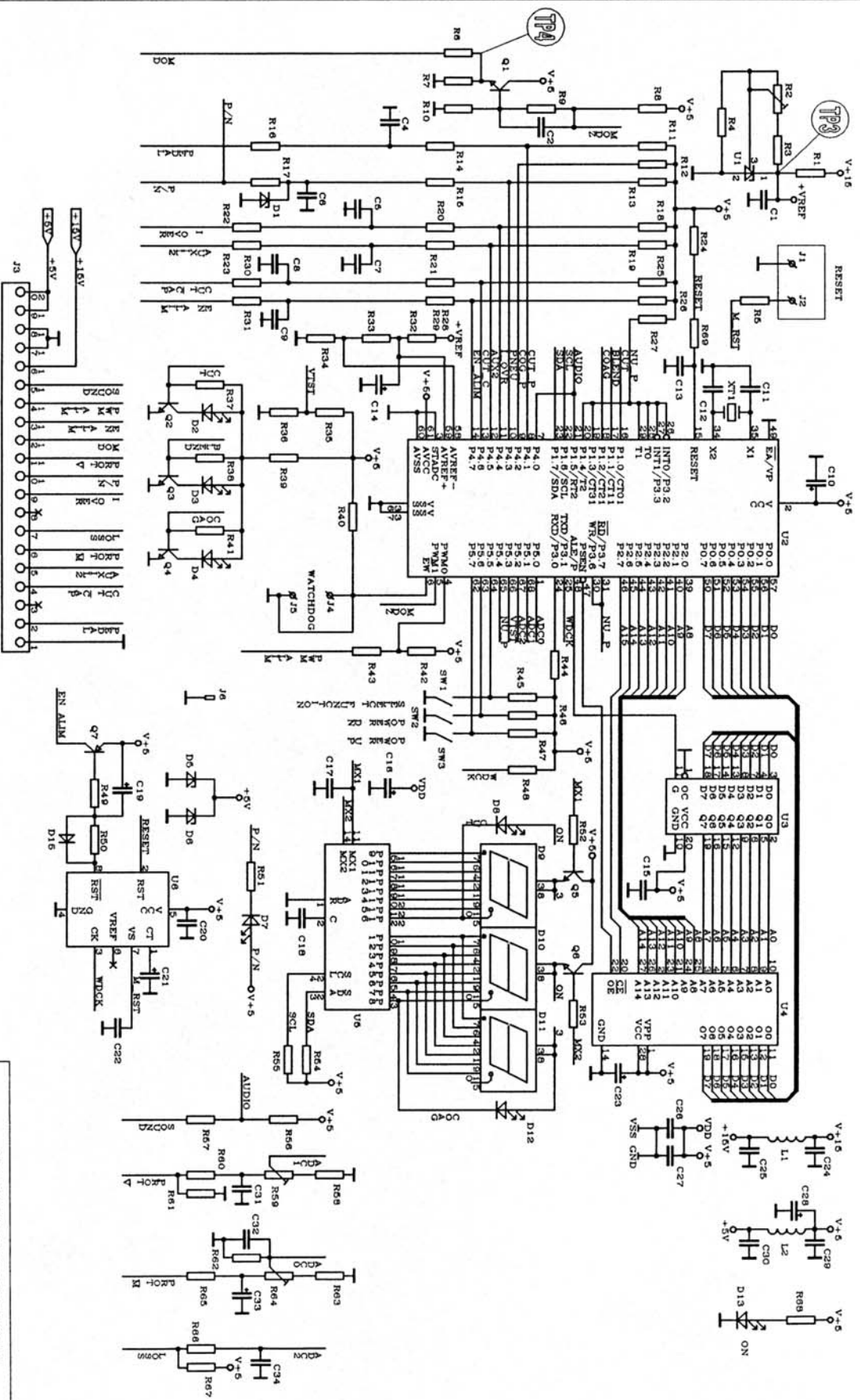
LEEP 1000 **801254**
Microcontroller Board
Reference **Part**

Bill Of Materials

Revised September 13, 1994

Revision 4A

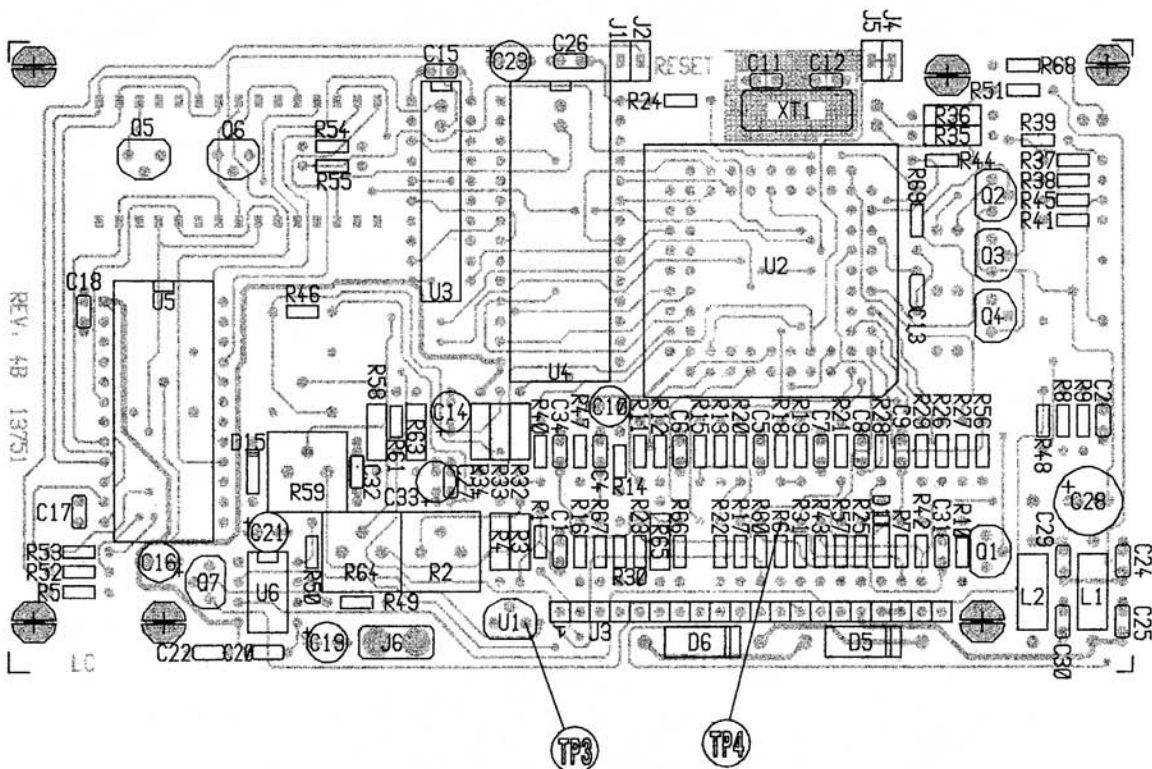
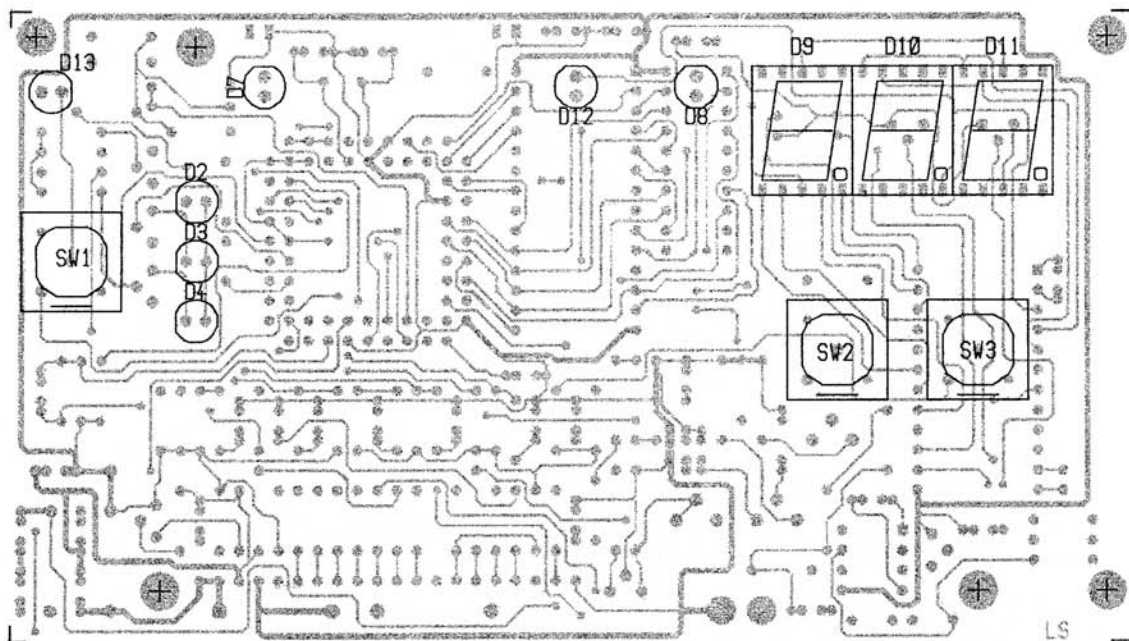
		Description	Code
C13	100n	Cermic Capacitor	400139
C14	10u/T/35V	Tantalum Electrolytic	400134
C15	100n	Capacitor Ceramic Capacitor	400139
C16	10u/T/35V	Tantalum Electrolytic Capacitor	400134
C17	1N/C	Ceramic Capacitor	400252
C18	2n2/C	Ceramic Capacitor	400256
C19	10u/T/35V	Tantalum Electrolytic Capacitor	400134
C20	100n	Ceramic Capacitor	400139
C21	1u/T/35V	Tantalum Electrolytic Capacitor	400173
C22	100n	Ceramic Capacitor	400139
C23	10u/T/35V	Tantalum Electrolytic Capacitor	400134
C24	100n	Ceramic Capacitor	400139
C25	100n	Ceramic Capacitor	420139
C26	100n	Ceramic Capacitor	420139
C27	100n	Ceramic Capacitor	420139
C28	47u/E/50V	Electrolytic Capacitor	420296
C29	100n	Ceramic Capacitor	420139
C30	100n	Ceramic Capacitor	420139
C31	10n	Ceramic Capacitor	420251
C32	100n	Ceramic Capacitor	420139
C33	1u/T/35V	Tantalum Electrolytic Capacitor	420173
C34	10n	Ceramic Capacitor	420251
D1	5V6/0.5W	Zener diode 5V6/0.5W	420018
D2	LED Y3	Yellow Led Lamp 3mm	420079
D3	LED Y3	Yellow Led Lamp 3mm	420079
D4	LED Y3	Yellow Led Lamp 3mm	420079
D5	5V6/5W	Zener diode 5V6/0.5W	420088
D6	5V6/5W	Zener diode 5V6/0.5W	420088
D7	GL5HR8	Red Led Lamp 5mm	420025
D8	GL5HY8	Yellow Led Lamp 5mm	420036
D9	HDSP5301	Common Anode 7 Segment Display	420080
D10	HDSP5301	Common Anode 7 Segment Display	420080
D11	HDSP5301	Common Anode 7 Segment Display	480080
D12	GL5HB8	Blue Led Lamp 5mm	420092
D13	NU	NOT USED	-----
D15	1N4148	Diode	480010
Q1	BC237B	NPN Transistor	427057
Q2	BC237B	NPN Transistor	427057
Q3	BC237B	NPN Transistor	427057
Q4	BC237B	NPN Transistor	427057
Q5	BC237B	NPN Transistor	427057
Q6	BC237B	NPN Transistor	427057
Q7	BC307B	PNP Transistor	427058
L1	100uH/275mA	Inductor	422006
L2	6uH8/800mA	Inductor	422007
U1	LM431	Voltage Regulator	482078
U2	80C552	Microcontroller	482077
U3	74HC373	74HC373	482066
U4	27256	EPROM256K	482067
U5	SAA1064	Display decoder	482080
U6	MB3773	Watch-dog timer	482088
XT1	12Mhz/Hc18	Quartz Crystal 12Mhz	252005
SW1	MTG_1241	Pushbutton SCHURTER	416094
SW2	MTG_1241	Pushbutton SCHURTER	416094
SW3	MTG_1241	Pushbutton SCHURTER	416094
J3	NU	NOT USED	-----
J6	399028	Connector Faston C.S.	399028



COOPERSURGICAL

17/7/95 LEEP SYSTEM 10004B

801282 - MICROCONTROLLER BOARD



COOPERSURGICAL		
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